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EDITORIAL

ADDICTION SSA

Cannabis and sport: A World Anti-Doping perspective

It is the opinion of the World Anti-Doping Agency (WADA) that cannabis should remain a banned substance in sport. The purpose of this editorial is to briefly summarize the elements of the discussion leading to the continued ban in sport.

Cannabis availability and its subsequent use, misuse and incidence of use disorder are increasing globally [1], even as the potency of the plant, gauged by delta-9-tetrahydrocannabinol (Δ^9-THC content), continues to increase [2]. It is not known what proportion of top-level athletes regularly consume cannabis. However, a recent review of self-reports that included more than 46000 athletes suggests that the proportion of athletes using cannabis is no different from the age-matched general population [3]. The World Anti-Doping Agency (WADA) banned cannabis in competition in all sports when the 2004 List of Prohibited Substances and Methods was first released under the World Anti-Doping Code [4]. It has since revisited and updated the cannabinoid category to include synthetic cannabinoids, increase reporting thresholds and to specifically exempt cannabidiol. In response to stakeholder requests, the WADA Executive Committee asked for a review of the status of cannabis. *Ad hoc* committees conducted a scientific, ethical and medical review of the effects of cannabis, focusing upon the primary psychoactive ingredient, Δ^9-THC, among the more than 100 distinct cannabinoids within the cannabis plant.

For any substance or method to be considered for inclusion in the Prohibited Substances List as per the World Anti-Doping Code, two of the following three equally-important criteria must be met: (1) it enhances or has the potential to enhance sport performance, (2) it represents an actual or potential risk to the health of the athlete and (3) it violates the spirit of the sport as defined in the Code.

In 2022, WADA reviewed more than 2700 articles for the present exercise. Inclusion of a scientific study or review to be further discussed was aided by adherence to scientific principles, where appropriate. This included application of rigorous methods, such as randomization of subjects to treatment groups, use of placebo controls and blinding of both subject and experimenter. Publication in a peer-reviewed scientific journal was required, and greater emphasis was placed upon studies that had results which were independently replicated. This enabled the Expert List, Athletic, Ethical and Health, Medical and Research committees to review the most current science and issue independent opinions on the subject. Finally, as part of the scientific review process, world leading experts on cannabis and addiction behavior were consulted to validate the conclusions of the committees.

Perspectives were mixed on the first List inclusion criterion. There were no rigorously conducted studies reviewed that indicated any direct ergogenic effect of cannabis. Nonetheless, a number of reports suggested that other benefits may be conferred by use of the drug, such as reduction in pain or anxiety [5, 6] associated with athletic competition. Some athletes report benefit following athletic performance, by facilitating recovery and reducing pain. Hence, while there may be benefits perceived by some athletes, there were no scientific data to support performance enhancement.

There was far more agreement on the second criterion—actual or potential risk to the health of the athlete. There is a comprehensive historical literature as well as a rapidly growing body of contemporary literature supporting the assertion that cannabis use can negatively impact the health, safety or wellbeing of the athlete. Acute intoxication can result in deficits in reaction time, temporal estimation and dexterity [7–11] as well as in psychiatric symptoms [12]. Not all studies detected similar changes; results will vary with the dose, route, subjects' experience and test methods employed. Chronic, frequent use of cannabis is associated with an increased risk of cannabis use disorder, as recently reviewed by Robinson and colleagues [13], as well as potentially other long-term changes [14, 15], including an increased risk of precipitation of severe mental illness [16]. There is a growing body of evidence that cannabis use can affect brain development and anatomy in adolescents and young adults [17–19], although the consequences of these changes are not yet firmly established. The potential of cannabis use to compromise the health and safety of the athlete qualifies the application of this criterion.

The third criterion, spirit of the sport, is considered as important as the other two criteria. The Ethics Advisory Group noted that spirit of sport encompasses a number of universal values of sport, and four aspects that were particularly relevant to the discussion on cannabis remaining in the list included the following:

- Excellence in performance: this could be undermined by consumption of cannabis during the in-competition period.
- Character and education: the role model aspect is not compatible with use of a substance that is still illegal in most parts of the world.
- Respect for rules and laws: its use violates the law in most countries in the world, in addition to Anti-Doping Organization rules in some instances.

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- Respect for self and other participants: the welfare and safety of other participants may be compromised by impaired judgment associated with the presence of cannabis in an athlete in-competition.

Based upon all the above considerations, which included input from the Ethics and Athletic committees at WADA, it was concluded that cannabis use met the Spirit of Sport criterion.

WADA emphasizes that prohibition of cannabis is in-competition only, which is defined as after 23:59 hours on the day prior to competition. The current decision limit of 180 ng/ml of Δ9-THC-COOH in urine and a cut-off of 150 ng/ml, plus the uncertainty of measurement of 30 ng/ml, takes this into account. Because of these high thresholds, primarily chronic, frequent cannabis users and athletes consuming high doses in-competition will be detected. Therefore, the cut-off generally will not affect the freedom of an athlete who wishes to legally consume cannabis outside of competition. Athletes who have a need for medicinal cannabis treatment should request a therapeutic use exemption (TUE).

KEYWORDS

Athlete, cannabis, doping, health, sport, WADA, Δ9-THC

AUTHOR CONTRIBUTIONS

Marilyn Huestis: Conceptualization (equal); writing—review and editing (equal). **Sabina Strano-Rossi:** Conceptualization (equal); data curation (equal); writing—review and editing (equal). **Yorck-Olaf Schumacher:** Conceptualization (equal); validation; writing—review and editing (equal). **Peter Harcourt:** Conceptualization (supporting). **Richard Budgett:** Conceptualization (equal); writing—review and editing (equal). **Mark Stuart:** Conceptualization (equal); writing—review and editing (equal). **Justice Tettey:** Conceptualization (equal); writing—review and editing (equal). **Irene Mazzoni:** Conceptualization (equal); data curation (equal); project administration (equal); resources (equal); writing—review and editing (equal). **Olivier Rabin:** Conceptualization (equal); supervision (equal); writing—review and editing (equal). **Anne Danion:** Conceptualization (equal); data curation (equal); project administration (equal); resources (supporting); writing—review and editing (equal). **Michael Culler:** Conceptualization (equal); writing—review and editing (equal). **David Handelsman:** Conceptualization (equal); writing—review and editing (equal). **Mario Thevis:** Conceptualization (equal); writing—review and editing (equal). **Audrey Kinahan:** Conceptualization (equal); project administration (equal); supervision (equal); writing—review and editing (equal).

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DECLARATION OF INTERESTS

No competing interests declared by any author.

DATA AVAILABILITY STATEMENT

Data were derived from public domain resources.

Thomas J. Hudzik[1]
Marilyn A. Huestis[2]
Sabina Strano Rossi[3]
Yorck-Olaf Schumacher[4]
Peter Harcourt[5]
Richard Budgett[6]
Mark Stuart[7]
Justice Tettey[8]
Irene Mazzoni[9]
Olivier Rabin[9]
Anne Danion[9]
Michael Culler[10]
David Handelsman[11]
Mario Thevis[12]
Audrey Kinahan[13]

[1]*ALA BioPharma Consulting, Monroe, NJ, USA*
[2]*Institute for Emerging Health Professions, Thomas Jefferson University, Philadelphia, PA, USA*
[3]*Department of Health Surveillance and Bioethics, Forensic Toxicology Laboratory, Università Cattolica del Sacro Cuore F. Policlinico Gemelli IRCCS, Rome, Italy*
[4]*Aspetar Orthopaedic and Sports Medicine Hospital, Doha, Qatar*
[5]*Australian Football League, Docklands, VIC, Australia*
[6]*Medical and Scientific Department, International Olympic Committee, Lausanne, Switzerland*
[7]*International Testing Agency—ITA, Lausanne, Switzerland*
[8]*United Nations Office on Drugs and Crime, Vienna International Centre, Vienna, Austria*
[9]*Science and Medicine Department, World Anti-Doping Agency, Montreal, Quebec, Canada*
[10]*Amolyt Pharma, Cambridge, MA, USA*
[11]*The ANZAC Research Institute Concord Repatriation General Hospital Gate 3, Concord, NSW, Australia*
[12]*Center for Preventive Doping Research, Institute of Biochemistry, German Sport University Cologne, Cologne, Germany*
[13]*Eirpharm, Ennis, County Clare, Ennis, Ireland*

Correspondence
Thomas J. Hudzik, ALA BioPharma Consulting, 170 Tournament Drive, Monroe, NJ, 08831 USA.
Email: tom@alabiopharma.com

ORCID

Thomas J. Hudzik https://orcid.org/0000-0002-3982-4475

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